Online Shoppers Intention Behavior Analysis

Project Type: Business Analytics Project
Tools Used: Python, Pandas, Matplotlib, Seaborn, Power BI
Dataset: Online Shoppers Purchasing Intention Dataset

Project Objective:

The objective of this project is to analyze online shoppers' browsing behavior and purchasing intention using data analytics and machine learning techniques. The project aims to identify important factors that influence customer engagement, purchase decisions, and conversion rates. Python is used for data cleaning and exploratory analysis, while Power BI is used to create an interactive dashboard and present meaningful business insights.

Tools and Technologies Used

Python – Data cleaning, exploratory data analysis and visualization
Pandas – Data manipulation and analysis
Matplotlib – Data visualization
Seaborn – Statistical visualization
Power BI – Interactive dashboard creation
CSV Dataset – Cleaned dataset used for analysis
Tools and Technologies Used
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Power BI – Interactive dashboard creation
CSV Dataset – Cleaned dataset used for analysis

Data Analysis Performed;

1 Univariate Analysis;

Website Type Distribution
Operating System Distribution
Administrative Duration

Explanation;

Univariate analysis was performed to understand the distribution of individual variables in the dataset. Website Type Distribution was analyzed to understand the types of websites visited by users. Operating System Distribution was analyzed to identify the operating systems used by visitors. Administrative Duration was analyzed to understand how much time users spent on administrative pages.

Bivariate Analysis:

Total Pages vs Total Duration
Bounce Rate vs Exit Rate
Revenue vs Non-Revenue Sessions
Exit Rate vs Revenue

Explanation:

Bivariate analysis was performed to understand the relationship between two variables. Total Pages and Total Duration were compared to identify the relationship between page visits and time spent on the website. Bounce Rate and Exit Rate were compared to understand visitor engagement. Revenue and non-revenue sessions were analyzed to understand purchasing behavior. Exit Rate and Revenue were also compared to identify whether exit behavior was associated with revenue generation.

Multivariate Analysis:

Visitor Type + Month
Administrative Duration + Product Related Duration by Revenue
Bounce Rate + Exit Rate by Revenue
Correlation Heat Map

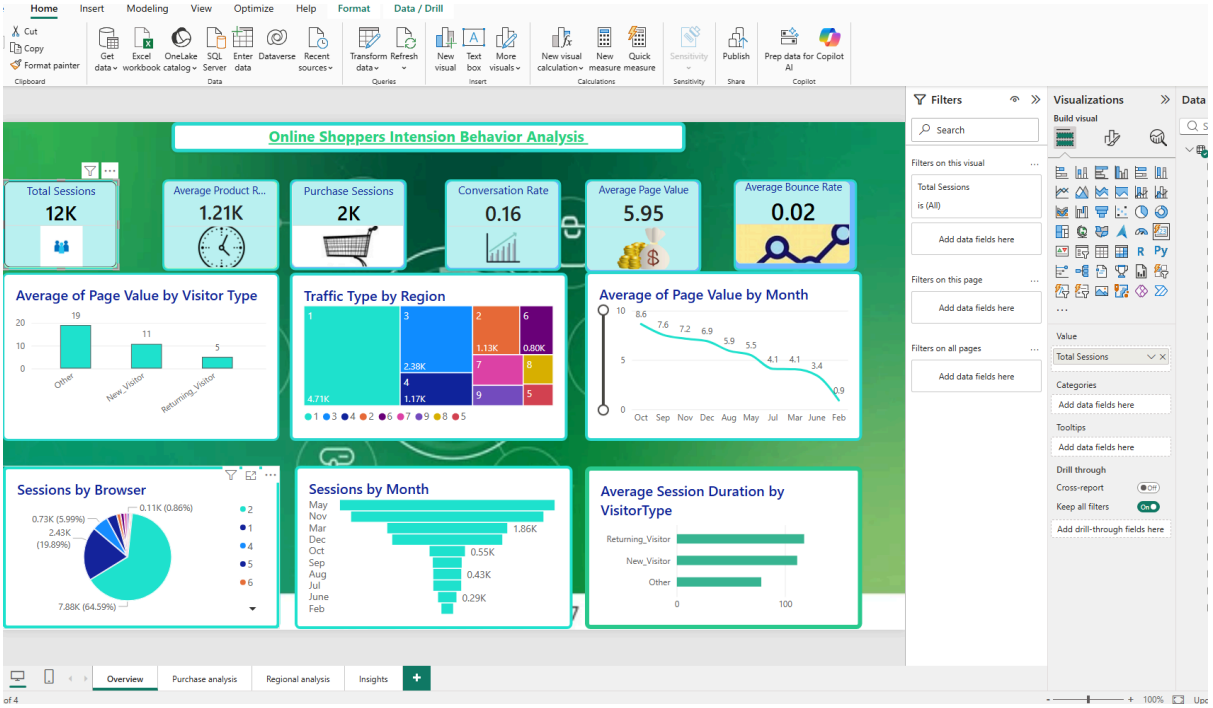
Explanation:

Multivariate analysis was used to analyze multiple variables together and identify more complex patterns in customer behavior. Visitor Type and Month were analyzed to understand purchasing behavior across different visitor categories and time periods. Administrative Duration and Product Related Duration were analyzed based on revenue generation. Bounce Rate and Exit Rate were analyzed by revenue status. A correlation heat map was also created to identify relationships between numerical variables.

Power BI Dashboard

The cleaned dataset was connected to Power BI to create interactive dashboards. The dashboards provide a visual representation of online shopper behavior, website engagement, visitor characteristics, session activity, page value, bounce rate, exit rate, and revenue-related metrics.

Dashboard Explanation:



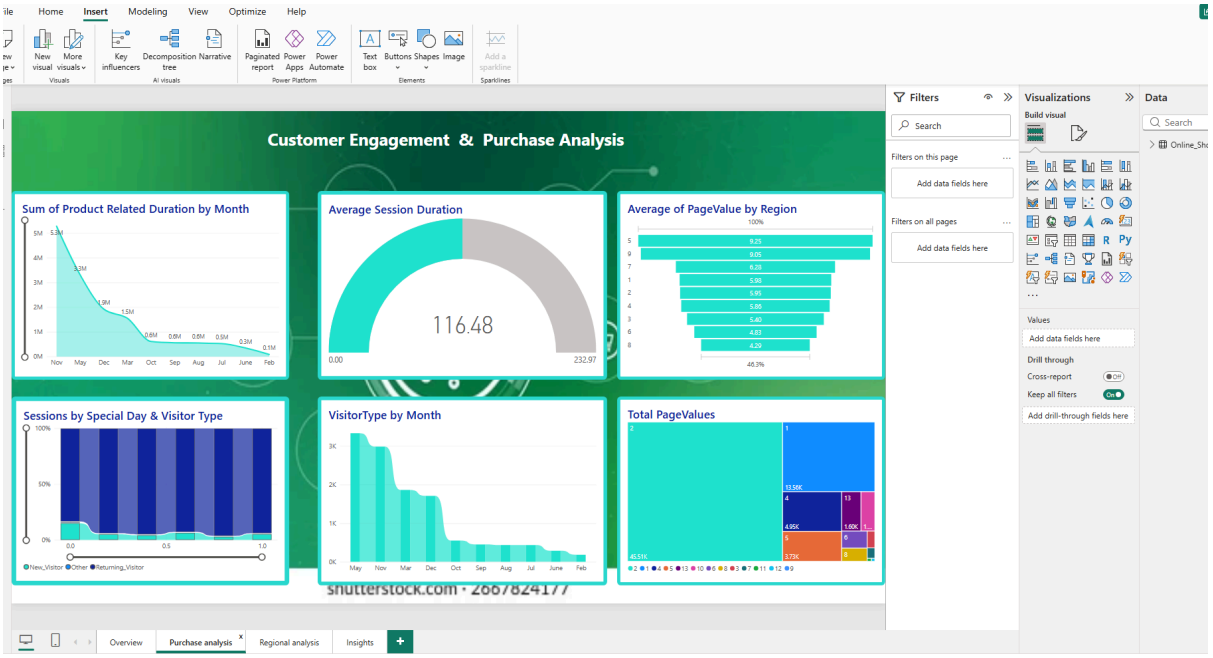
Dashboard Page 1 – Overall Online Shopper Performance

Total Sessions
Average Product Related Duration
Purchase Sessions
Conversion / Purchase Rate
Average Page Value
Average Bounce Rate
Average Page Value by Visitor Type
Traffic Type by Region
Average Page Value by Month
Sessions by Browser
Sessions by Month
Average Session Duration by Visitor Type

<u>Page 1 Description:</u>

Dashboard Page 1 focuses on the overall performance and engagement of online shoppers. It presents key performance indicators such as total sessions, purchase sessions, conversion rate, average product-related duration, page value and bounce rate. The dashboard also provides insights into visitor type, traffic type, monthly trends and browser usage.

Dashboard Page 2 – Customer Engagement and Purchase Analysis

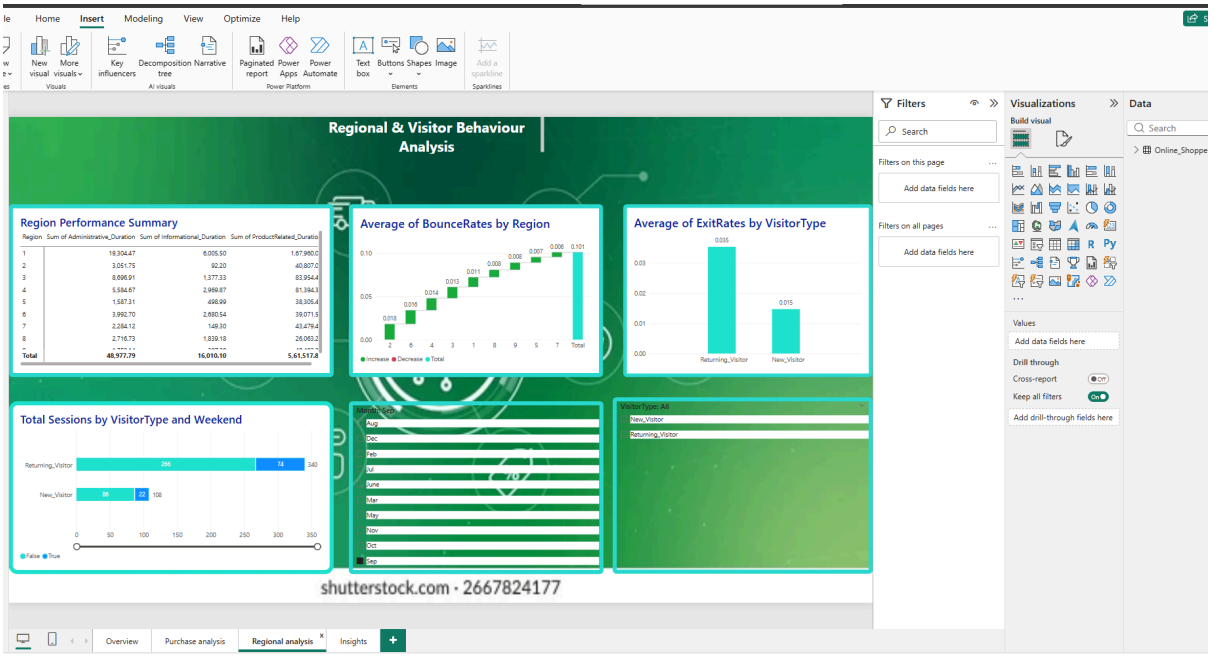


Average Bounce Rate by Region
Conversion / Purchase Rate
Sessions by Month
Average Exit Rate by Visitor Type
Average Session Duration by Month
Traffic Type by Visitor Type

<u>Page 2 Description:</u>

Dashboard Page 2 focuses on customer engagement and purchasing behaviour. It helps to understand how customers interact with the website through bounce rate, exit rate, session duration, monthly sessions and conversion behaviour. The visuals also help compare customer behaviour across different visitor and traffic type

Dashboard Page 3 – Regional and Visitor Behaviour Analysis



Region Performance Summary
Average Bounce Rate by Region
Average Exit Rate by Visitor Type
Total Sessions by Visitor Type and Weekend
Visitor Type Analysis
Traffic Type / Regional Distribution

<u>Page 3 Description:</u>

Dashboard Page 3 focuses on regional performance and visitor behaviour. It compares customer engagement across different regions and visitor types. The dashboard helps identify differences in bounce rate, exit rate, sessions and purchasing behaviour across customer segments.

Findings / Key Insights:

Key Insight 1 – Visitor Behavior

Visitor behavior varies across different visitor types. New visitors and returning visitors show different levels of website engagement and purchasing activity.

Key Insight 2 – Session Engagement

The relationship between total pages viewed and total session duration indicates that website engagement is associated with the amount of time users spend browsing the website.

Key Insight 3 – Bounce and Exit Rate

Bounce rate and exit rate provide important indicators of website engagement. Higher values may indicate that visitors are leaving the website without exploring more pages.

Key Insight 4 – Revenue and User Behavior

Revenue-generating sessions show different behavioral patterns compared with non-revenue sessions. Page value, browsing activity and product-related duration can be used to understand purchasing intention.

Key Insight 5 – Monthly Pattern

The analysis shows that shopper activity and page value vary across months, indicating that customer engagement and purchasing behavior are not uniform throughout the year

Business Recommendations:

1. Improve Website Engagement

The business should improve website content and navigation to encourage visitors to explore more pages and spend more time on the website.

2. Reduce Bounce and Exit Rates

Pages with high bounce or exit rates should be reviewed to identify usability, content or navigation issues.

3. Focus on High-Value Visitors

Visitor segments showing higher engagement and purchasing activity can be targeted with personalized offers and relevant product ***recommendations.***

4. Optimize High-Performing Periods

Monthly trends can be used to identify periods of higher customer engagement and plan promotional campaigns accordingly.

5. Improve Product Page Experience

Since product-related browsing behavior is associated with purchasing intention, product pages should provide clear information, attractive presentation and easy navigation.

Final Story

The analysis provides an overall view of online shopper behavior and purchasing intention. Python was used to explore individual variables, relationships between variables and complex behavioral patterns. The results were then presented through Power BI dashboards to make the analysis easier to understand. The dashboards highlight website engagement, visitor behavior, session activity, page value, bounce rate, exit rate and revenue-related patterns. These insights can help businesses understand

customer behavior, improve website performance and support data-driven decision-making.

Conclusion:

This project demonstrates how data analytics and visualization can be used to understand online shopper behavior. The analysis identified important patterns related to visitor type, website engagement, session duration, page activity, bounce rate, exit rate and revenue. Python helped perform detailed exploratory analysis, while Power BI provided an interactive way to communicate the findings. The combined analysis can support businesses in improving customer engagement, website performance and purchasing conversion.